# SIMATS ENGINEERING
## ASSESSMENT TOOL 3: Mock Interview

**COURSE NAME:** Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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**COURSE OUTCOME COVERED**
**CO5:** *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 25**
Question Count: 10

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**Code of Conduct**
I **M. Meghana(192525435)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: *M. Meghana*

**Assessment Tasks**
**Mock Interview**
Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

**Mock Interview Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Technical Knowledge | 25% | Shows deep and accurate technical understanding | Good technical understanding | Basic understanding | Weak understanding |
| Problem Solving | 20% | Responds with strong architecture-level reasoning | Reasonable reasoning | Basic reasoning | Weak reasoning |
| Communication | 20% | Answers are confident, precise, and professional | Mostly clear communication | Moderate clarity | Weak communication |
| Practical Awareness | 20% | Connects concepts to real-world implementation well | Some practical relevance | Limited practical relevance | No practical linkage |
| Confidence and Professionalism | 15% | Highly confident and professional | Generally professional | Moderately professional | Low confidence and professionalism |

## 1. Explain the difference between HDFS and traditional file systems.

HDFS (Hadoop Distributed File System) is designed to store and process very large datasets across multiple computers. It divides files into blocks and distributes them across Data Nodes. It also keeps multiple copies of blocks for fault tolerance. Traditional file systems generally store data on a single system or storage device.

Main advantages of HDFS:

- Distributed storage
- High scalability
- Fault tolerance through replication
- High-throughput data access

## 2.Why is replication important in distributed storage systems?

Replication means maintaining multiple copies of data on different nodes. It is important because hardware failures are common in distributed systems. If one node fails, another node can provide the required data.

**Benefits:**

- Prevents data loss
- Provides high availability
- Improves fault tolerance
- Supports faster recovery

For example, HDFS commonly uses a replication factor of **3**, meaning three copies of a data block can be maintained.

## 3.How does Map Reduce divide work between map and reduce phases?

Map Reduce processes large datasets in two main phases.

1. **Map Phase:** Input data is divided into smaller parts. Mapper processes each part and produces key-value pairs.
2. **Shuffle and Sort:** Intermediate key-value pairs are grouped according to their keys.
3. **Reduce Phase:** Reducers process the grouped data and generate the final output.

**Flow:**
`Input → Map → Shuffle & Sort → Reduce → Output`

This approach allows large datasets to be processed in parallel.

### 4.What role does YARN play in large-scale Hadoop processing?

YARN stands for **Yet Another Resource Negotiator**. It manages resources and schedules jobs in a Hadoop cluster.

Its main components are:

- **Resource Manager:** Manages overall cluster resources.
- **Node Manager:** Manages resources on individual nodes.
- **Application Master:** Manages the execution of a particular application.

YARN allocates CPU and memory efficiently and allows multiple applications to run on the same Hadoop cluster.

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### 6. What is the purpose of ETL in a big data pipeline?

ETL stands for **Extract, Transform, and Load**. It is used to collect data from different sources and prepare it for analysis.

- **Extract:** Collect data from databases, files, applications, etc.
- **Transform:** Clean, filter, validate, and convert the data.
- **Load:** Store the processed data in a target system.

**Flow:**

```
Data Sources → Extract → Transform → Load → Analytics
```

ETL improves data quality and makes data suitable for reporting and analysis.

## 7. How does Apache NiFi help in data integration workflows?

Apache NiFi is a data integration and flow management tool. It helps organizations move and process data between different systems.

**Functions of NiFi:**

- Collects data from multiple sources.
- Routes and filters data.
- Transforms data into required formats.
- Connects databases, APIs, files, and other systems.
- Provides data provenance and monitoring.
- Supports error handling and retries.

Thus, NiFi makes data integration automated and easier to manage.

## 8.What are the key failure points in a Hadoop cluster and how would you handle them?

The major failure points in Hadoop are Data Nodes, Name Node, Resource Manager, disks, and network connections**.**

**Handling methods:**

- **Data Node failure:** Use data replication.
- **Name Node failure:** Use Name Node High Availability.
- **Resource Manager failure:** Use YARN High Availability.
- **Disk failure:** Monitor disks and maintain replicated data.
- **Network failure:** Use redundant network connections.

Regular monitoring and fault-tolerant architecture help maintain cluster availability.

## 9. How would you design a secure data ingestion pipeline for a large organization?

A secure data ingestion pipeline can be designed as:

**Data Sources → Secure Ingestion → Validation → Storage → Processing → Analytics**

Security measures include:

1. Use encryption for data in transit and at rest.
2. Authenticate users and data sources.
3. Apply role-based access control.

4. Validate incoming data.
5. Maintain audit logs.
6. Protect sensitive information.
7. Monitor the pipeline for unauthorized access.
8. Use secure network connections.

This ensures that data remains confidential, accurate, and protected throughout the pipeline.

**10.How would you explain a scalable Hadoop-based processing system to a non-technical manager?**

I would explain Hadoop using the example of a large task divided among many workers. Instead of one computer processing all the data, Hadoop distributes the data and processing across multiple computers.

- **HDFS** stores data across multiple machines.
- **YARN** manages computer resources.
- **Map Reduce** divides large processing tasks into smaller tasks.
- **Replication** protects data from hardware failures.

If the amount of data increases, additional machines can be added to the cluster. Therefore, Hadoop provides scalability, fault tolerance, and efficient processing of big data.